



# Coding with Claude Code

The disciplined engineering loop for computational-materials research: seven phases, the constraint theory underneath them, and one units bug that only a human caught.

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■ TITLE & HOOK · NOT VIBE-CODING · ~5 MIN

## **Coding with Claude Code, for people who validate the physics**

- Not vibe-coding: the disciplined loop that produces code you can trust
- Most 'AI for coding' is written for web apps
- Workflow transfers cleanly; vocabulary doesn't
- You're fluent in the science; the new layer is the engineering workflow
- One running example, two threads, meeting at one number

## Who this is for

- Comp-materials scientists & grad students who run atomistic/multiscale sims
- KMC, MD/statics, DFT, and write the code that does it
- Given: fluency in the science
- New: the AI-assisted engineering workflow on top
- Explicitly not for vibe coders

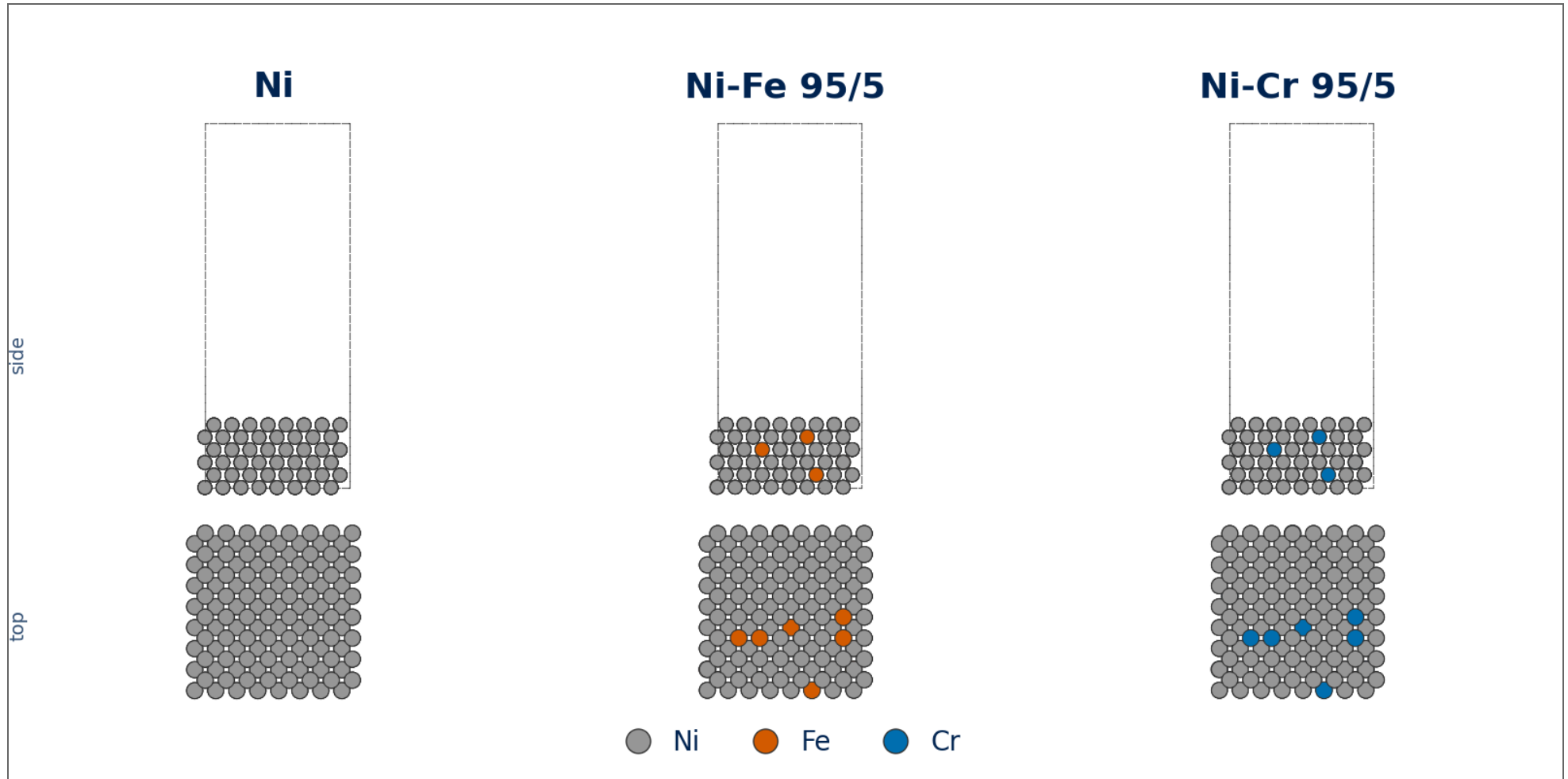
## The two threads that converge

- Thread 1, substrate: periodic\_system\_generator → Ni/Ni-Fe/Ni-Cr FCC(001) slabs, vacancies, step edges → LAMMPS+EAM minimize → pyKMC
- Thread 2, correctness: ~23-commit harmonic-TST (Vineyard) prefactor campaign in pyKMC
- They meet at the Ni(100) surface-hop barrier dE
- Same dE the slabs study = same dE that pins the units regression test
- That convergence is why this is THE example

**ANCHOR** Thread 1 slab generation (Ni/Ni-Fe 95/5/Ni-Cr 95/5 FCC(001)) + Thread 2 HTST prefactor work, meeting at the Ni(100) surface-hop barrier dE (EAM ~0.63 eV, Mehl 1999; DFT ~0.82–0.86 eV, exchange-dominated).

■ THE PICTURE · THREAD 1 SUBSTRATE

**The substrate, built from code: Ni / Ni-Fe / Ni-Cr FCC(001) slabs**



■ THE REFRAME · DEV-SPEAK  $\rightleftharpoons$  RESEARCH · ~7 MIN

## **The Rosetta stone: one substitution drives everything**

- Deliverable: behaviour a user sees → a physical claim that survives scrutiny
- Every translation is a consequence of that one substitution
- PRD = study/method spec (a mini pre-registration)
- Bug = wrong PHYSICS: no crash, green suite
- QA = validating the physics, not click-testing a UI

## The unit of work, translated

- Idea → a physics question, an artifact, or a code capability
- Feature → a new physical capability or measured quantity (the Vineyard backend)
- Bug → wrong physics, often green-and-silent (nu0 in Hz where the clock wants  $\text{ps}^{-1}$ )
- Refactor → restructure the solver without changing the numbers
- Sprint → a simulation campaign (the ~23-commit HTST push)

**ANCHOR** Idea: 'give pyKMC physically-correct per-event rate prefactors instead of one fixed constant.' Feature: `pykmc/rate_constant/backends/htst.py`. Bug: nu0 Hz vs  $\text{ps}^{-1}$ , every rate  $10^{12}$  too large, silently.

## Artifacts, agents, and what NOT to carry over

- PRD: spec the END STATE, let implementation move underneath
- research.md asset: a sprint-lived cache allowed to rot
- Memory-less agent: why knowledge lives in contracts, not your head
- Trigger surface: the always-on cost of a skill, minimize it
- Don't import: 'ship fast, user won't notice' (our failure mode is silently-wrong)

**ANCHOR** PRD example: 'Deliver D\_vac(Ni,500K) with an Arrhenius barrier validated against an EAM-NEB value; archive trajkmc.xyz + reference\_table.pickle.' RateConstantConfig is the typed end-state contract.

## The dictionary: the unit of work

| term               | in software development                                           | in materials research                                                                |
|--------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Deliverable</b> | Behaviour a user sees: a shipped, working feature in the product. | A physical claim, or a reusable capability, that survives scrutiny.                  |
| <b>Feature</b>     | New behaviour a user can see and use.                             | A new physical capability or a measured quantity (the Vineyard / HTST rate backend). |
| <b>Bug</b>         | Code that crashes or returns visibly wrong output.                | Wrong PHYSICS, often with no crash and a green test suite: the silent killer.        |
| <b>Refactor</b>    | Restructure code without changing its observable behaviour.       | Restructure a solver or driver without changing the numbers it produces.             |

| term          | in software development                                     | in materials research                                                                       |
|---------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Sprint</b> | A fixed-length iteration delivering a slice of the backlog. | A simulation campaign, or a focused push to land one capability (the ~23-commit HTST push). |

## The dictionary: plan and board

| term                                       | in software development                                                 | in materials research                                                                                                    |
|--------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>PRD (product requirements document)</b> | The spec of what to build and why: problem, users, acceptance criteria. | A study or method spec, a mini pre-registration: system, conditions, the observable with acceptance bands, output files. |
| <b>Ticket / Jira card</b>                  | One scoped, trackable unit of work on a board.                          | One scoped sub-task of a study or code change ("minimize this slab", "add the units conversion").                        |
| <b>Kanban board</b>                        | Tickets flowing through columns: to-do, doing, done.                    | The dependency graph of a study: stages in series, seeds / temperatures / compositions in parallel.                      |
| <b>Vertical slice / tracer bullet</b>      | One thin end-to-end path through every layer, working early.            | One end-to-end short run touching every pipeline stage (build → minimize → search → select → analyze).                   |

| term                         | in software development                               | in materials research                                                                                      |
|------------------------------|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>research.md<br/>asset</b> | A scratch notes file for a task, allowed to go stale. | A throwaway cache of method details (parameters, units, citations) scoped to one campaign, allowed to rot. |

## The dictionary: the agent and its context

| term                                           | in software development                                                                | in materials research                                                                                                                      |
|------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Token</b>                                   | The unit an LLM bills and "thinks" in: text encoded to integer ids.                    | A metered second compute budget; structurally, one catalog entry (a Reference Event index in pyKMC).                                       |
| <b>Context budget / smart zone / dumb zone</b> | The model's usable working memory; quality degrades as it fills.                       | Effective context far shorter than the advertised window (~100K usable is a heuristic); the dumb zone manufactures silently-wrong physics. |
| <b>Compaction vs handoff</b>                   | Compaction summarizes the chat to continue; handoff starts a fresh session from a doc. | Compaction buys continuity (lossy by omission you don't control); handoff buys purity (a disposable pointer doc for one task).             |
| <b>Memory-less agent</b>                       | A fresh agent with no memory between tasks; state must be re-read.                     | Why durable method knowledge lives in contracts and docs, not your head: the                                                               |

| term                                            | in software development                                                                               | in materials research                                                                                                                     |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 |                                                                                                       | next agent starts blank by design.                                                                                                        |
| <b>Ralph loop / coding-agent execution loop</b> | A dumb bash loop re-running one agent against a prompt and memory file until done (Geoffrey Huntley). | An unattended loop over independent runs (run_overnight.sh under nohup): one run per iteration, append-only memory, per-iteration commit. |

## The dictionary: quality and design

| term                                | in software development                                                                | in materials research                                                                                                             |
|-------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>QA / click-testing a UI</b>      | Checking the product behaves: click the UI, run the tests.                             | Validating the physics: barrier vs NEB, detailed balance, MSD linearity, units, one imaginary mode.                               |
| <b>TDD / regression test</b>        | Write a failing test first; keep tests green to guard behaviour.                       | A test that pins a PHYSICAL contract and must stay red until the physics is right (a correctness guarantee, not a design driver). |
| <b>Green and wrong</b>              | A passing test suite that does not actually prove correctness.                         | A passing suite that has ENCODED the physics error and defends it (where the units bug lived).                                    |
| <b>Deep module / thin interface</b> | A small interface over a large implementation (Ousterhout); the best modules are deep. | The same taste applied to solvers: the seam you wrap a test around, so deep modules lead to strong feedback loops.                |

| term                                   | in software development                                                                       | in materials research                                                                                     |
|----------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Skill /<br/>trigger<br/>surface</b> | A reusable, auto-triggering capability; its always-on description is a standing context cost. | Same idea: minimize the always-on description, preferring a dispatched sub-agent or a one-paragraph rule. |

■ FOUNDATIONS · WHY THE WORKFLOW HAS THIS SHAPE · ~9 MIN

## **The constraints that force the procedure**

- Smart zone / dumb zone: effective context is far shorter than the advertised window
- A bigger window doesn't move that limit, it just ships more dumb zone
- Feedback loops are the ceiling, not the model, not the prompt
- Tokens are a metered second compute budget, and mirror the KMC engine
- A procedure you don't understand is one you'll abandon when rushed

## Smart zone / dumb zone: a correctness control

- Cost and quality are two facts, not one: the quadratic doesn't cause the degradation
- Cost: attention is  $O(n^2)$  in length (compute and memory), so big contexts are expensive
- Quality: positional bias degrades use well before the window fills (Lost in the Middle, TACL 2024; RULER)
- '~100K' is a model-specific heuristic, not a measured constant
- For us the dumb zone manufactures silently-wrong physics: size to the smart zone, clear don't compact

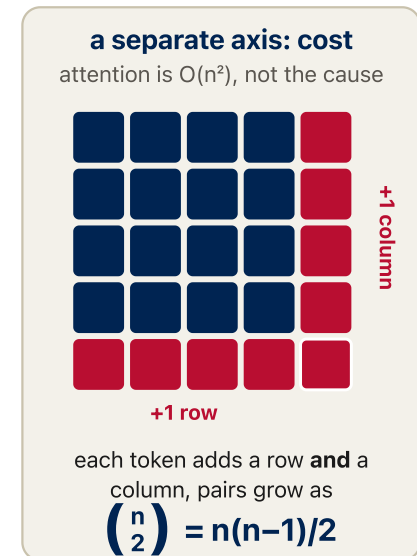
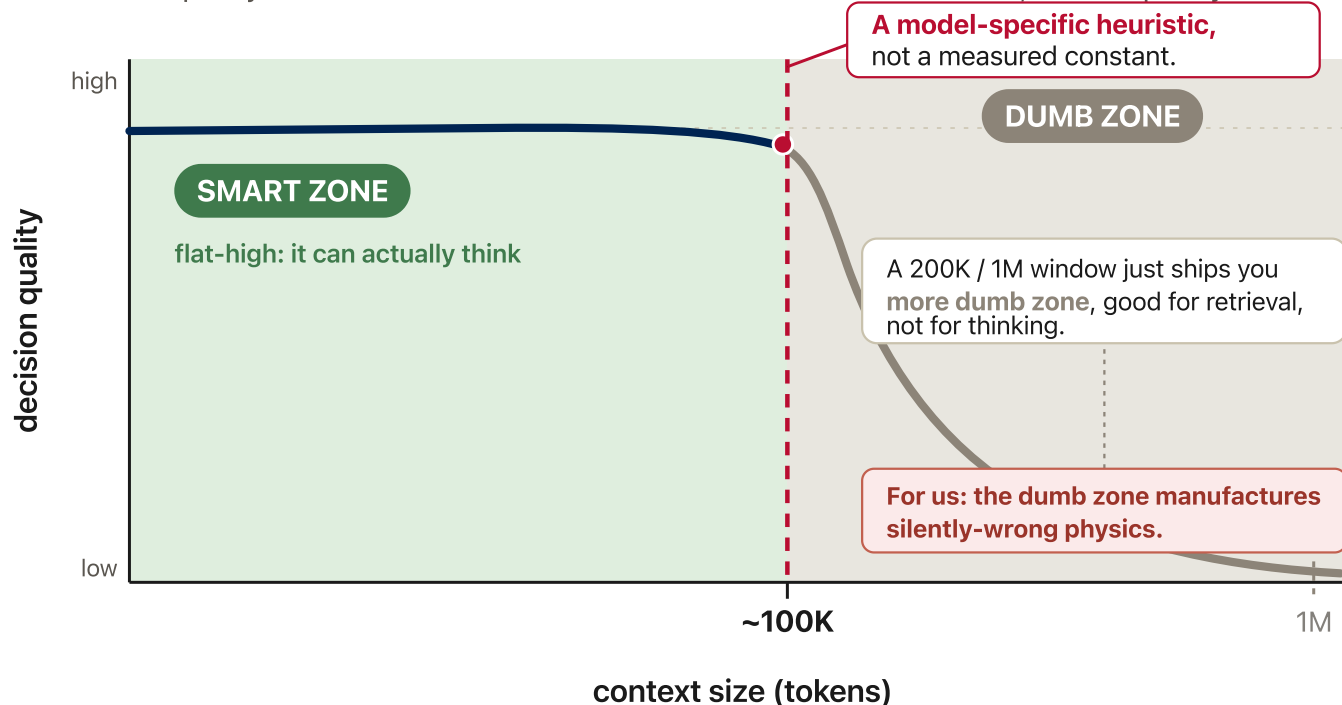
**ANCHOR** pykmc/htst/constants.py: omega\_eV\_to\_hz() and hz\_to\_per\_ps() are two functions because they are two facts, the fine-grained correctness the agent holds in the smart zone and drops in the dumb zone.

## ■ THE PICTURE

# Smart zone / dumb zone: a correctness control

### Usable attention degrades before the advertised window

Decision quality vs. context size: the smart zone is a correctness control, not a capacity number



Quality dip = positional bias (Liu et al., "Lost in the Middle", TACL 2024; RULER), a separate phenomenon from the  $O(n^2)$  cost.

**Sizing a task to fit the smart zone is a correctness control.**

Clear, don't compact: start each unit of work from a fresh, small context.

## Feedback loops are the ceiling, and ours can be green-and-wrong

- Web feedback: types, unit tests, lint, a screenshot
- Ours adds a second tier, the physics: detailed balance, NEB match, MSD linearity, one imaginary mode, ~THz prefactor
- 'Green and wrong': a passing suite that has ENCODED the error and defends it
- Deep modules → testable → strong loops; a bad codebase makes a bad agent
- Ousterhout is skeptical of strict TDD: our regression test pins a physical contract, it doesn't 'drive design'

**ANCHOR** HTST\_Test\_Report/run\_htst\_ab.py: paired constant-vs-htst runs on matched seeds, built to fail if HTST yields no finite  $\nu_0$ , a physics-tier feedback loop. RateConstantConfig.require\_one\_negative\_mode encodes physics in the type. A sibling on-lattice build emitted a  $\nu_0$  of 470 THz that compiled clean: see the case study.

## Tokens are the currency, and the KMC engine in miniature

- Tokens: a metered second budget, billed per call, input priced apart from output
- Feed a summary contract plus a file path, never the multi-MB dump
- `reference_table.pickle` IS a learned vocabulary; the KMC loop IS encode/compute/decode
- `n_current` = vocabulary hit (free); `n_new` = miss (`nsearch=20` ARTn search)
- Decode silently corrupts when the number carries a different unit than the consumer assumes

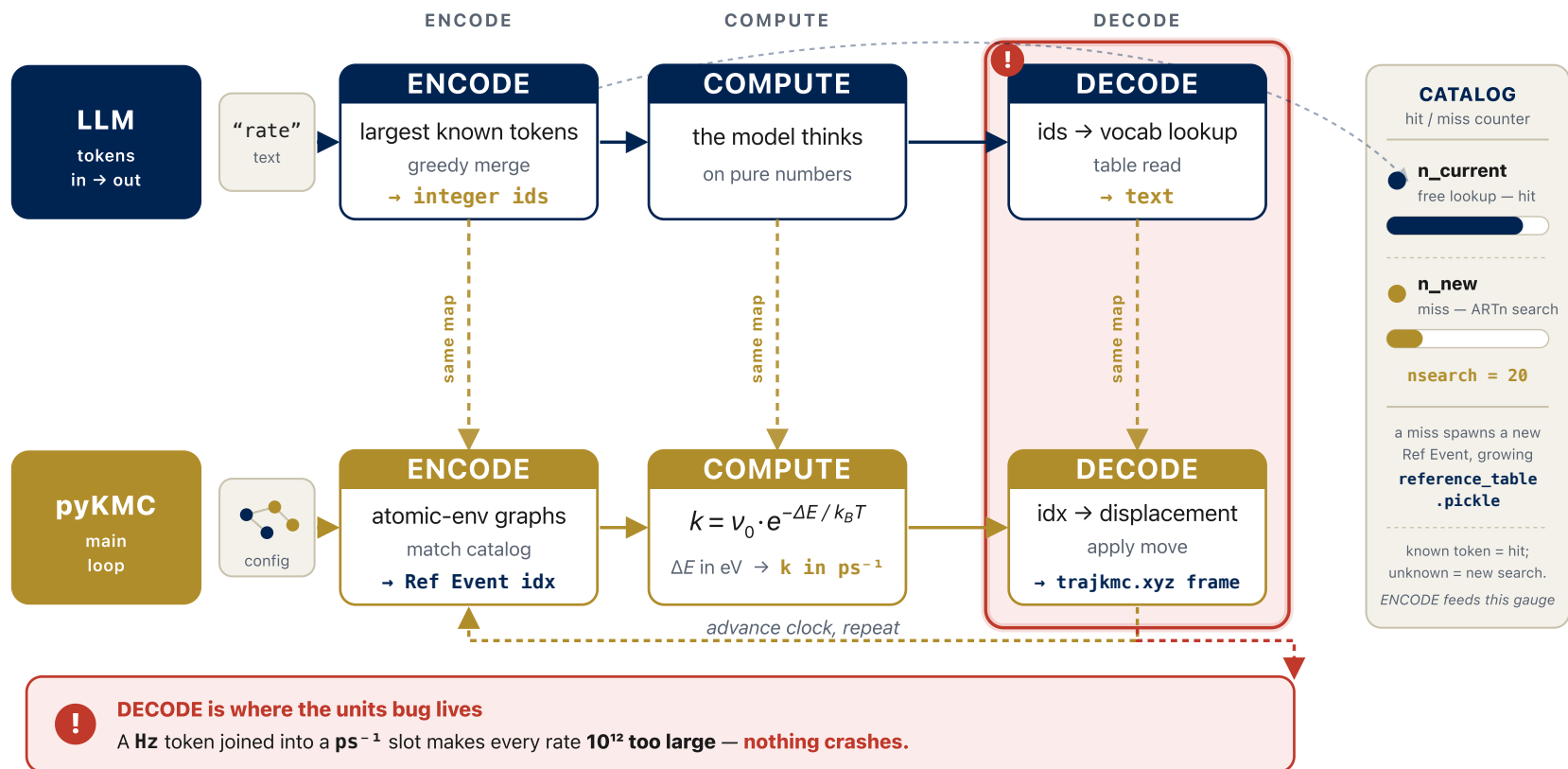
**ANCHOR** validation\_basin: trajkmc.xyz >1MB, reference\_table.pickle ~5.7MB, lammmps.log.0 ~11MB. The Ref Event is a token id; the Hz→ps<sup>-1</sup> bug is a decode failure.

## ■ THE PICTURE

# Tokens are the currency, and the KMC engine in miniature

### Same machine, twice over

An LLM token pipeline laid directly on the pyKMC main loop — the units bug lives in **DECODE**



■ THE SEVEN PHASES · THE SPINE OF THE METHOD · ~11 MIN

**Idea → Research → Prototype → PRD → Kanban → Execution → QA**

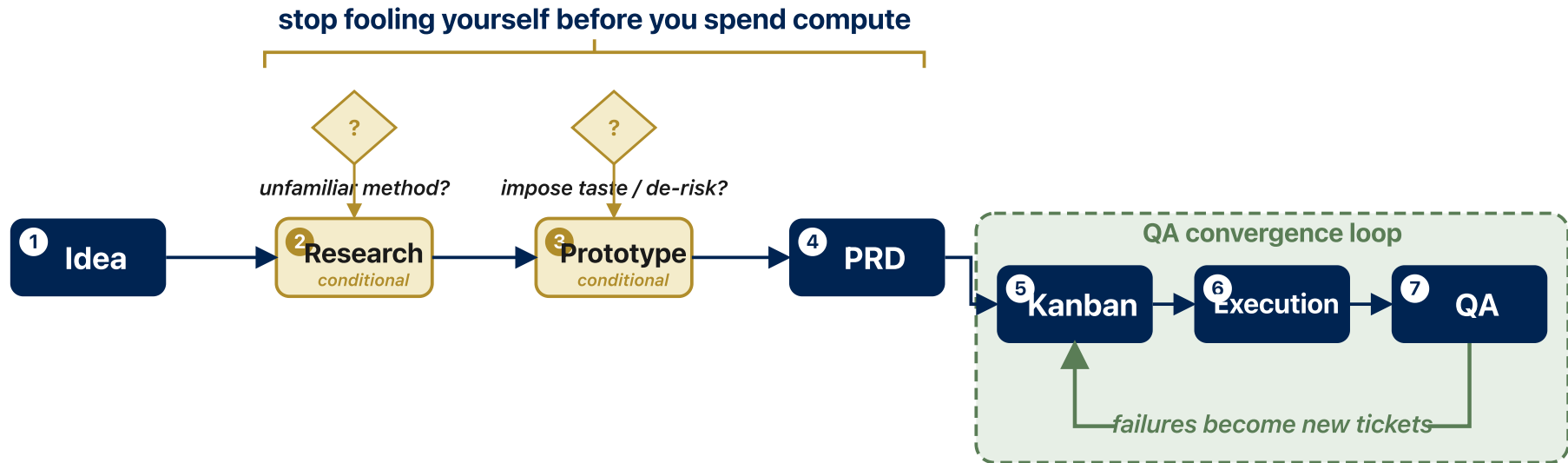
- Two conditional phases: Research (2) and Prototype (3), skipped when small/familiar
- Last three (Kanban→Execution→QA) form a loop until the science converges
- QA failures become new tickets → back to Kanban
- A map, not a gate: tiny work skips 2, 3, most of 4
- Value: names for the steps, so nobody silently drops QA

## ■ THE PICTURE

# The shape: a linear front half, a converging back-half loop

### The seven-phase loop

Linear front half, two conditional phases, and a QA convergence loop



#### MATERIALS MEANING

- 1 Idea** sharp dissatisfaction or a target number
- 2 Research** cache method details; stamp build / date
- 3 Prototype** sweep knobs till the catalog stabilizes
- 4 PRD** `RateConstantConfig` + **DO NOT MIX** units header

- 5 Kanban** dependency graph; cut vertically
  - 6 Execution** build launches real compute (`mpirun` / `srun`)
  - 7 QA** a human reads units & magnitudes
- green tests are not validated physics*

## ■ ONE LINE PER PHASE

### The seven phases, in a simulation campaign

| # | Phase                      | What it means for a study                                                                     |
|---|----------------------------|-----------------------------------------------------------------------------------------------|
| 1 | Idea                       | A physics question, an artifact, or a code capability. State a sharp dissatisfaction or a...  |
| 2 | Research<br>(conditional)  | Cache method details you'll forget: parameter semantics, the units convention, failure...     |
| 3 | Prototype<br>(conditional) | A throwaway convergence or recipe shakedown: sweep knobs on a tiny cell until the catalog...  |
| 4 | PRD                        | A study/method spec, a mini pre-registration: system, conditions, observable + acceptance...  |
| 5 | Kanban                     | The dependency graph of the study: stages in series (build→minimize→search→select→analyze)... |
| 6 | Execution                  | Same loop, but 'build' launches real compute (mpirun/srun), genuinely asynchronous. One...    |
| 7 | QA                         | Validating the physics. The agent-producible half is real (unit tests, type-check, lint...    |

## Phase 1–3: Idea, Research, Prototype

- Idea: state a sharp dissatisfaction or a target number, not a solution
- 'Rates use a constant prefactor and that's wrong', not 'add a function vineyard()'
- Research (conditional): cache method details into a sprint-lived research.md, stamp with conditions
- Profiling harness, not a notebook: eskm  $\sim 2.5\text{--}6\times$  faster than Python-FD, agreement 0.014%
- Prototype (conditional): sweep knobs on a tiny cell, commit the winner, delete the losers

**ANCHOR** Phase 2: commit d52f541 profiling harness (eskm dynamical\_matrix vs Python-FD) → benchmarks/htst\_prefactor/RESULTS.md. Phase 3: in.minimize\_slab\_Ni\_\*.Imp variants; the NiAlH\_jea default failed for Ni-Cr, FeNiCrCoAl-heaweight.setfl won.

## Phase 4: PRD, spec the end state, grill the design

- PRD = study/method spec: system, conditions, observable + acceptance bands, output files
- NOT the implementation: over-pinned code rots against fast-moving scripts
- Grill the design: which potential? which seeds? what defines convergence? what's the null result?
- `require_one_negative_mode` encodes the physics in the type
- Record deferred scope: RpaBackend bare-Vineyard until the Sharia-Henkelman  $\kappa$  lands (deferred, not broken)

**ANCHOR** `RateConstantConfig(style: constant|htst|rpa, free_radius, fd_step, nu0_min/max_THz, require_one_negative_mode)`. vineyard.py header: 'Units convention (DO NOT MIX) ... nu0 output: Hz (linear, NOT angular)'. Review commit a752002 (Hugo Moison).

## ■ THE CODE

### Phase 4: PRD, spec the end state, grill the design

```
1 RateConstantConfig(  
2     style = "constant" | "htst" | "rpa", # thin interface literal  
3     free_radius = ...,  
4     fd_step = ...,  
5     nu0_min_THz = ..., nu0_max_THz = ...,  
6     require_one_negative_mode = True, # physics invariant AT the  
    interface:  
7 ) # a true first-order saddle has  
8     # exactly one imaginary mode.
```

## Phase 5: Kanban, vertical slices, tickets are worktrees

- Board = dependency graph: stages in series, seeds/temps/compositions in parallel
- First ticket is a tracer bullet: one thin slice through every stage
- Cut vertically (build→minimize→search→select→analyze), never horizontally
- Tickets are literally worktree + branch names
- Independent fixes (freeze-core, units) ride parallel branches

**ANCHOR** HTST chain: 9327519 base → Vineyard+config+PHONON install → manager-ops → batched compute → reference backfill → active/refined backfill → recycling skip. Worktrees: pyKMC-htst-work, pyKMC-freezecore-work (fix/freeze-core-pbc), pyKMC-htst-units-work (fix/htst-rate-units).

■ PHASE 6 ZOOMED IN · THE EXECUTION LOOP · ~7 MIN

## **Ralph: a dumb bash loop that beats orchestration, and you already own one**

- Backstop, one feature per iteration, append-only memory, per-iteration commit
- You already run it: `run_overnight.sh` under `nohup`
- 'done' must be runnable physics, not a green test suite
- Backstop is wall-clock + queue depth, not just iterations
- `set -e` vs `set -uo pipefail` IS the AFK-vs-attended decision

## The 'done' predicate must be runnable physics

- Web: agent flips passes:true on green type-check + tests (defensible there)
- Physics: a green suite tells you the code RAN, never that the physics is RIGHT
- passes = a machine-checkable acceptance band in eV/Å/ps, run by the real solver
- Predicate 1: require\_one\_negative\_mode read off the actual dynamical matrix
- Predicate 2: finite-nu0 A/B gate; coverage<1.0 is a logged WARNING, not a pass

**ANCHOR** run\_htst\_ab.py: status='passed' only if the failures list (built from physics observables) is empty. A 511-atom Si smoke run hit 33.3% nu0 coverage, logged as partial, not a benchmark.

## ■ THE CODE

### The 'done' predicate must be runnable physics

```
1 # run_htst_ab.py: status == "passed" only if failures is empty
2 if n_events <= 0:
3     failures.append("no reference events")
4 if "nu0" not in htst_table.columns:
5     failures.append("HTST table has no nu0 column")
6 if n_finite_nu0 <= 0:
7     failures.append("HTST produced no finite nu0 values")
8 if nu0_coverage < 1.0:
9     warnings.append(...) # WARNING, not a failure
```

## Memory, sizing, commits, and the role shift

- Append, never update: a memory-less agent reconstructs state from the progress file
- Record the physics state: 'coverage 0.71; three events fell back to k0'
- One feature per iteration: compute\_prefactors\_batch shards the giant Hessian
- Per-iteration commit = bisectable; the surgical one-line units fix d705932 in its own worktree
- Ralph demotes you from anal-retentive planner to requirements-gatherer

**ANCHOR** Commit 82e7793 compute\_prefactors\_batch (batch size chosen by the d52f541 profiling harness); c9a74ad never-recompute-recycled = idempotency; chain ec6aa5f→...→c9a74ad bisectable.

■ CONTEXT & HANDOFF · SURVIVING MANY FRESH AGENTS · ~4 MIN

## **Compaction buys continuity; handoff buys purity**

- Grep the convergence line, not the log: 30 tokens vs 30,000
- Compaction is lossy by omission you don't control: drops the load-bearing caveat
- Handoff: a disposable temp-dir pointer doc carrying one task's slice
- Point, not copy: references stay correct as targets evolve
- Portable markdown → adversarial physics review by a different agent

## The DIY sub-agent round trip

- Hand off a hard-to-judge bit to a prototype session (separate budget)
- It does the disposable work, returns a compressed verdict
- RESULTS.md IS the handoff-back doc, stamped with its conditions
- Same shape as a dispatched sub-agent, zero machinery
- The real dispatched ones: apps routes five AGENT.md agents (analysis, viewer, classify, blender, schematic); pylatkmc, three (codegen, ingest, validation)
- Promote durable facts to CONTEXT.md; the handoff is the courier, not the archive

**ANCHOR** Commit d52f541 round trip: the eskm-vs-Python-FD question needed a measurement; benchmarks/htst\_prefactor/RESULTS.md returned the verdict (eskm faster, conditions attached) and the planner chose the eskm path. Each AGENT.md is scoped by a description that also says what it must NOT touch.

■ PRACTICE & HONEST LIMITS · DESCEND OR SKIP · ~8 MIN

## **What the spine leaves out: clusters, reproducibility, failure modes, objections**

- HPC reality: SLURM on the Alliance (DRAC), and the sandbox is the security boundary
- A reproducible artifact, not a reproducible agent (many AI projects don't even run)
- The units bug is one genre: a taxonomy of LLM failure modes
- The four objections you will get, answered
- Case study: the on-lattice prefactor build, and the 470 THz the AI couldn't see was wrong
- All of these are descend-or-skip depth; the seven-phase spine is the talk

## ■ DRILL-DOWN · HPC

### Running it for real: SLURM on the Alliance (DRAC)

- Submit and watch: sbatch, salloc, squeue/sq, sacct, scancel
- Always required: --time and --account (plus --mem / --mem-per-cpu, --gpus-per-node)
- \$SLURM\_TMPDIR is fast node-local scratch, deleted at job end; module load for the stack
- /home (small, backed up) vs /scratch (big, purged) vs /project (shared); Globus for big transfers; one array job, not 1000 sbatch calls
- Security: never an unattended Ralph with --dangerously-skip-permissions on a login node or with live credentials; the sandbox is the boundary

**ANCHOR** A study run = an array job over seeds/temps, each staging inputs into \$SLURM\_TMPDIR, launching pyKMC under srun, copying results back to /project before the node is wiped. The Ralph loop runs inside a container/VM, never with live cluster credentials.

## ■ DRILL-DOWN · REPRODUCIBILITY

### **The artifact is reproducible even though the agent is not**

- Agent runs are non-deterministic; AI projects often don't even execute (~31.7% fail out of the box)
- The reproducible unit is NOT the agent transcript
- It is: committed code + pinned env (container/lockfile) + fixed RNG seed + the regression test
- Declared vs runtime dependencies drifted ~13.5x in that study: pin them
- Same discipline as any KMC study: fix the seed, archive the inputs, pin the physical contract

**ANCHOR** Reproducible study = fixed RNG seed for the BKL draws + pinned EAM potential + archived reference\_table.pickle + the Ni(100) dE regression test. Rerunnable next year regardless of which agent wrote it. pylatkmc goes further: its codegen is byte-deterministic (PYTHONHASHSEED-proof), so the generated C re-emits identically. (arXiv:2512.22387)

## ■ DRILL-DOWN · FAILURE MODES

### **The units bug is one genre: a taxonomy**

- Slopsquatting: ~19.7% of recommended packages don't exist (USENIX Security 2025); a fabricated import is a supply-chain hole
- Insecure code: ~45% of samples carried an OWASP Top-10 vulnerability (Veracode 2025)
- Higher defect density: AI pull requests ~1.7x the issues of human ones (CodeRabbit 2025)
- Domain silent killers: wrong constants, off-by-one lattice indexing, green-but-wrong tests
- The units bug is the last row; no linter catches it, only a human reading the physics

**ANCHOR** The Hz to  $\text{ps}^{-1}$  bug is the 'green-but-wrong test' genre: it ran, the suite was green, the numbers were  $10^{12}$  off. The taxonomy says the units bug was not bad luck, it was a predictable category. The case study's 470 THz prefactor is the same genre in a different codebase: the proof it's a category.

## ■ DRILL-DOWN · OBJECTIONS

### **Objections, answered**

- "Won't this deskill students?" AI is the tactical programmer; the human stays the strategic designer, and reviews every diff
- "How do I trust code I didn't write?" You trust the regression tests that pin physical contracts, not the provenance
- "How is a stochastic run reproducible?" It isn't; the pinned artifact is
- "Does this fit open science / FAIR?" Yes: version prompts and CLAUDE.md, disclose AI use, share the artifact
- Disclosure is becoming a norm (Ten Simple Rules for AI-Assisted Coding in Science, arXiv:2510.22254)

## ■ DRILL-DOWN · FIGURES AS CODE

### **Figures are code, not AI images**

- AI image generators are unreliable for technical schematics: wrong lattices, missing components, broken connections (SciFlow-Bench)
- One controlled study found 99.8% of generated images contained fabricated detail
- So generate every figure with code: ASE, matplotlib, Graphviz, TikZ, or hand-authored SVG
- The figure becomes a version-controlled, re-runnable artifact: the deck's own thesis
- Every figure here has a source in assets/src or is hand-authored SVG; none is AI-generated

**ANCHOR** assets/src/gen\_arrhenius\_units\_bug.py and gen\_ni\_slabs.py generate the Arrhenius and slab figures (the slab reads the real validation\_basin structure); the flow diagrams are hand-authored SVG. Re-run when the physics changes.

## Case study: making the AI compute an attempt frequency

$$\nu_0 = \frac{\prod_i \omega_i^{\text{init}}}{\prod_i \omega_i^{\text{sad}}} \cdot \frac{1}{2\pi\hbar}$$

- $\nu_0$  is the pre-exponential in  $k = \nu_0 \cdot \exp(-E_a/kT)$ : a ratio of normal-mode frequencies, minimum over saddle
- Three scripts: `hessian_lammps.py` (ASE+LAMMPS Hessian) → `kappa_rpa.py` (modes,  $\nu_0$ ,  $\kappa$ ) → `canonical_kappa.py` (NEB+Dimer saddle)
- Dispatched to one scoped agent, `ingest-htst-bridge`: catalogue and prefactors only, never codegen
- ~Two weeks; the agent wrote nearly all of it. Then the numbers came back.

**ANCHOR** `pylatkmc/ingest/htst: vineyard_prefactor()` in `kappa_rpa.py`;  
`canonical_kappa.py` drives NEB+Dimer → mass-weighted Hessian →  $\nu_0$  for  
`surface_1NN_inplane` and `subsurface_1NN_inplane`.

## It compiled, ran, and emitted a plausible number

- Subsurface  $\nu_0$  came back at 470 THz: physically impossible for a 1NN hop
- The tell:  $\nu_0$  GREW as I added free atoms (167  $\rightarrow$  470 THz,  $r=6 \rightarrow r=8$ ): an artefact, not physics
- Cause: surface atoms  $\sim 0.88$  Å above the buried vacancy rattle at  $\sim 1$  meV and inflate the Vineyard ratio
- Separately, a real saddle once reported '0 negative modes': the imaginary mode was mis-read as a zero mode
- Every one of these passed the code's own internal checks; the mode counts balanced

**ANCHOR** family\_prefactors.csv (subsurface): "was 167-470 THz at  $r=6/r=8$  with full free region ... soft-mode contamination." Nothing crashed; nothing was red.

## What caught it: physical judgment, not a green suite

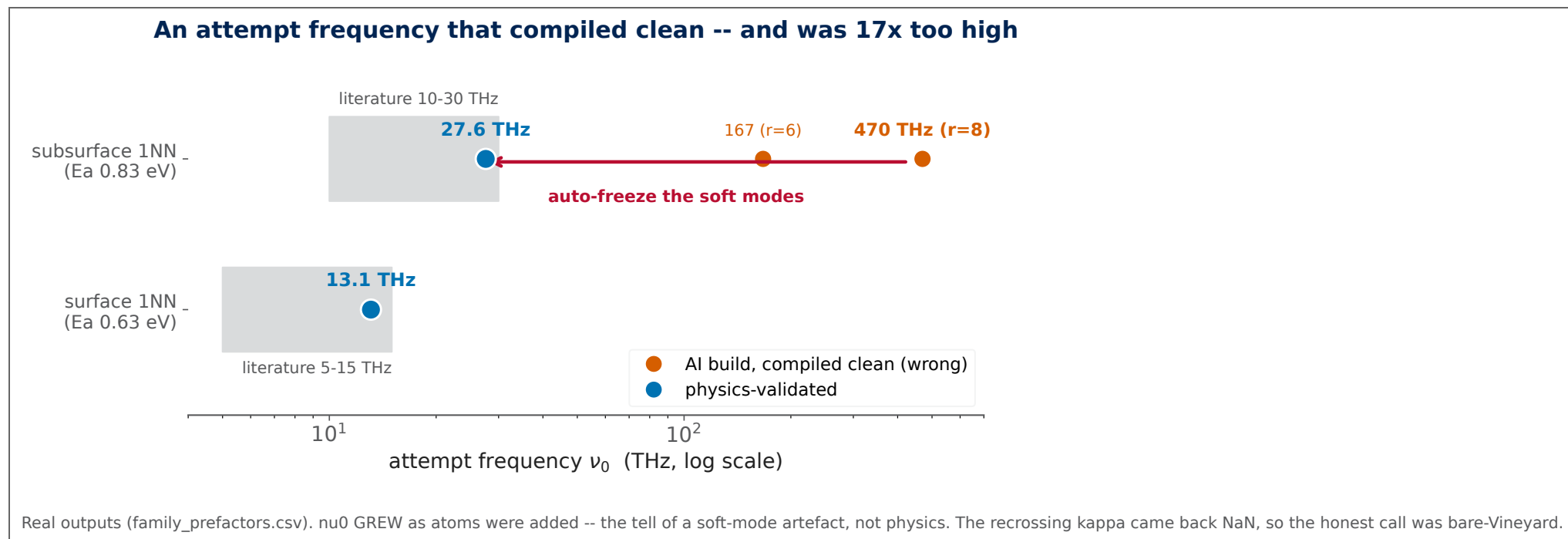
- Literature is the yardstick: FCC 1NN  $\nu_0$  is 5–30 THz, full stop; 470 fails on sight
- Cross-check two  $\nu_0$  paths ( $n_{\text{zero\_modes}}$  3 for the slab vs 0 for a frozen cluster): divergence flags the bug
- $\kappa$  recrossing returned NaN (perturbation theory unreliable): the honest call was bare-Vineyard, logged

```
1 # kappa_rpa.py: refuse to report an unreliable recrossing factor
2 if alpha_f * kT > perturbation_safety:      # default 0.5
3     warn("RPA perturbation theory unreliable")
4     return float("nan"), diagnostics
```

**ANCHOR** Each catch became a regression test; the soft-mode fix is one line: `free_min_z = mover_z + 0.88` in `canonical_kappa.py` auto-freezes the rattling surface.

## ■ THE PICTURE · CASE STUDY · 4 OF 4

**The lesson: the agent built it; only the physics knew it was wrong**



**ANCHOR** Validated  $\nu_0$ : surface 13.1 THz ( $\times 1.31$ ), subsurface 27.6 THz ( $\times 2.76$ ). No linter, type-checker, or unit test can tell you 470 THz is unphysical. That check is irreducibly the human's job.

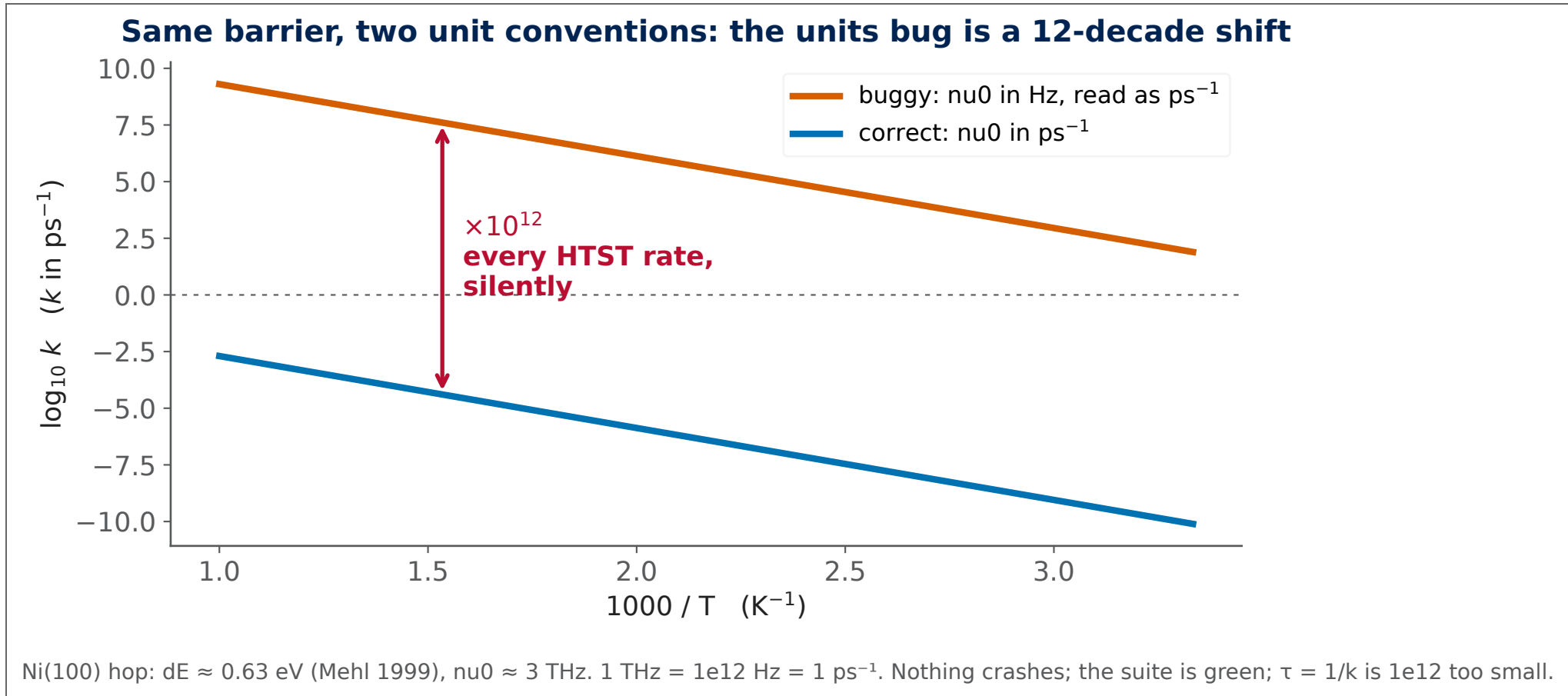
## ■ QA, TAKEAWAYS & ATTRIBUTION · ~2 MIN

### **Phase 7: the units bug, and the one thing to remember**

- nu0 emitted in Hz (~1–10 THz), consumed as  $\text{ps}^{-1}$  → every rate  $10^{12}$  too large
- Nothing crashed; the suite was green; the tests had pinned the error
- Fix d705932: one-line hz\_to\_per\_ps() at the backend boundary
- Regression d6b921f: red by  $10^{12}$  before, green after, at the Ni(100) dE where both threads meet
- QA = validating the physics. Green tests are not validated physics.

## ■ THE PICTURE · THE UNITS BUG

The units bug is a 12-decade shift, plotted from the pinned numbers



## Attribution & the open questions

- Structure: five Matt Pocock talks; materials mapping is original to this collection
- Documents skills you may already run: grilling, to-prd, to-issues, tdd
- Open: merge Research + Prototype into one 'de-risk the method' phase?
- Open: keep the name 'Ralph' or 'coding-agent execution loop'?
- First cut: the seven-phase mapping is a proposal for discussion, not a settled standard



Speaker notes

